

Session 6.3

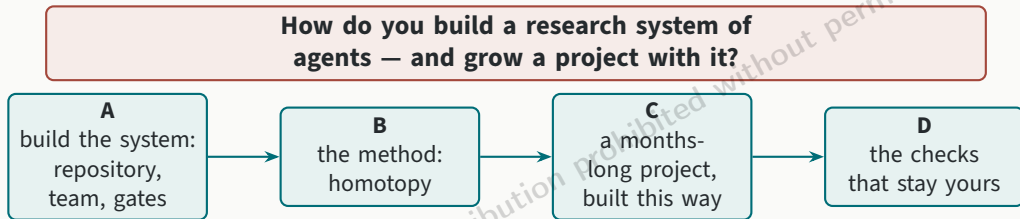
How to Build With Agentic AI: The System, the Homotopy Method, a Research Case, and the Checks

Part three of five: set up a research system of agents you can trust — the repository, the team, the gates — then grow a project with it one validated margin at a time, as a months-long research project did



Zhigang Feng

The Question, and the Route



- ▶ Deck 6.2 showed how the machine works. This deck is what you build around it, and how you work with it.
- ▶ Two scales of one discipline, both from the instructor's work: a twenty-two-minute task and a months-long research project.
- ▶ **The system is set up once; the method is used every day; the checks never go away.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1–§4.3.

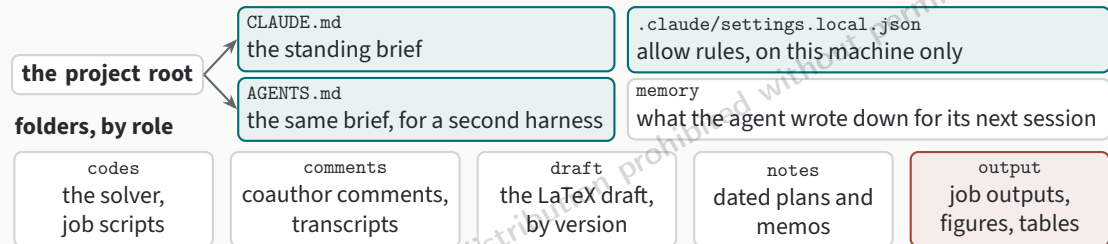
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Build the System

The repository, the team and the gates: what you write before the first prompt



The Repository Is the Agent's Memory



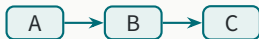
every arrow of Deck 6.1's pipeline starts in one of these folders and ends in another; the two files at the top are what let a session that knows nothing pick the chain up

- ▶ The national-debt project's root, by role. Your home folder holds your defaults; the repository holds what a coauthor clones, and where they disagree the project wins.
- ▶ **The harness reads these files first at every start, so a session that knows nothing picks the work up from them.**

Source: Instructor's project repository, read 23 Sep 2026; folder names by role; no content shown.

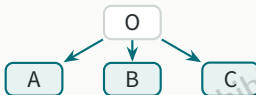
Design the Team: Three Wirings, and Who Checks Whom

sequential



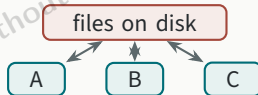
each list seeded
by the last output

hierarchical



workers report up;
only O sees all

blackboard



all read and
write one store

- ▶ Strip the vocabulary away and multi-agent designs differ in one respect: who writes into whose list.
- ▶ Two names for every good: who makes it, who checks it. The producer is never the only reviewer; a review panel is the parallel wiring, its reports kept apart.
- ▶ **In a research project the blackboard is files on disk — which is why a file-driven workflow is the one that survives a restart.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3–§4.4.

One Agent or a Team? Marginal Value Against Marginal Tokens



- ▶ One vendor's own eval: a lead agent with workers beat a single agent by 90.2%, and multi-agent runs use about $15\times$ the tokens of a chat.
- ▶ The quota is tokens per minute, not agents; context binds per worker; and coordination — reading and reconciling n outputs — usually binds first.
- ▶ **Reach for more workers when the task is genuinely wide, not to feel parallel.**

Source: Anthropic engineering, multi-agent research systems (13 June 2025); Miao & Feng, *AI for Economists*, ch. 15, §4.3.

A Hook That Never Fires, and the Rule That Works

never fires

```
"PreToolUse": [  
  { "matcher": "Write(data/*.csv)",  
    "hooks": [{ "type": "command",  
                  "command": "check_schema.sh" }]}]
```

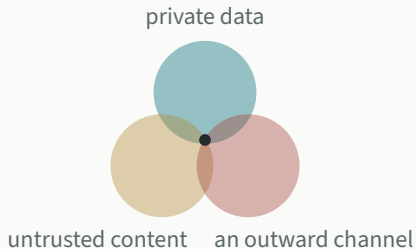
works

```
"PreToolUse": [  
  { "matcher": "Write|Edit",  
    "if": "Write(data/*.csv)",  
    "hooks": [{ "type": "command",  
                  "command": "check_schema.sh" }]}]
```

- ▶ A **hook** is a command the harness runs at a named event; before a tool call it can refuse the call, so it is a gate. The left one fails silently: a matcher with punctuation is read as a regular expression against the *tool name*, and no tool is named `Writedata`.
- ▶ Two fields, two jobs: the matcher selects on the tool's *name*; the `if` field filters on its *arguments*, in permission-rule syntax, and is only evaluated on tool events.
- ▶ **Test a gate by trying to break the rule it enforces. A rule you have never seen fire is a rule you have not got.**

Source: Claude Code hooks reference, matcher and condition fields, read 23 Sep 2026; the failure was observed on the course machine on 12 September 2026.

Least Privilege, Three Named Risks, and the Lethal Trifecta



“If your agent combines these three features, an attacker can **easily trick it** into accessing your private data and sending it to that attacker.”

- ▶ **Least privilege**: no write for a read-only task, no network for an offline one; match the grant to the task, not the model.
- ▶ The risk list: *prompt injection* first, *sensitive information disclosure* next; *excessive agency* climbed it in 2026.
- ▶ Anything the agent reads (e-mails, web pages, retrieved documents, even tool descriptions) is an untrusted injection channel.
- ▶ Data can reach the model's instructions: with sensitive data, cut the outward channel.

Source: OWASP Gen AI Security Project, Top 10 for LLM applications (2025; 2026 revision) and Top 10 for Agentic Applications (Dec 2025); S. Willison, “The lethal trifecta for AI agents”, 16 June 2025.

Initialise a Research Repository in Eight Steps

step	the command, or the file	what it buys
1 a folder under version control	<code>mkdir savings-model && cd savings-model && git init</code>	one command back
2 the standing brief	<code>CLAUDE.md</code> : goal, three rules, one gate	the spec, re-read every start
3 least privilege	<code>.claude/settings.local.json</code> : an allow list of read and git commands	the blast radius, written down
4 one skill	<code>.claude/skills/weighted-rate/SKILL.md</code>	the weights rule, as an asset
5 one reviewer	<code>.claude/agents/reviewer.md</code> : reads, reports, never edits	a checker who is not the producer
6 one gate	a <code>PreToolUse</code> hook with the <code>if</code> field on data writes	a rule that refuses, not reports
7 the checkpoint	<code>RESUME.md</code> : state, and the next step	the next session starts here
8 the first commit	<code>git add -A && git commit -m "initialise"</code>	the tree a coauthor clones

Six files and two commands: a collaborator who clones it gets your agent, not your habits. Each skill or agent file needs a description that says when to use it.

Source: Course machine, 23 Sep 2026; Miao & Feng, *AI for Economists*, ch. 15, §3.3; Claude Code hooks reference, read 23 Sep 2026.

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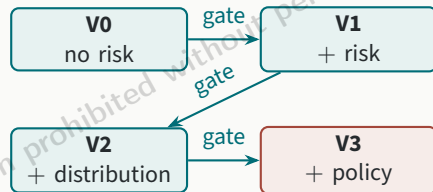
The Method: Homotopy

Solve the easy version first, validate it, then deform it toward the research task



The Homotopy Idea: Solve the Easy Version First

- ▶ Start from a version with clear intuition and a benchmark you understand.
- ▶ Use that validated version as the starting point for the next one.
- ▶ Add one margin at a time, and attribute a new bug to the margin you just added.
- ▶ **Never ask the agent to solve the full problem in one jump.**

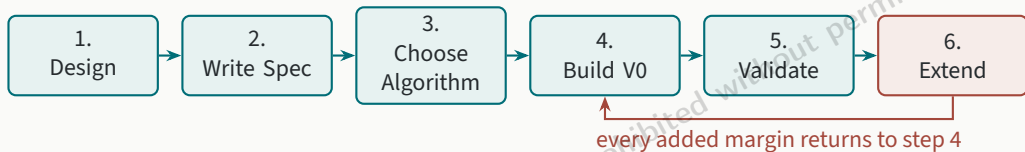


each step small, each step validated;
a failed step returns to the last version
that passed

Why the name: homotopy continuation starts from a problem with a known solution and deforms it continuously toward the target. V0 to V3 are gates along that deformation.

Source: Judd, *Numerical Methods in Economics* (1998), ch. 5; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

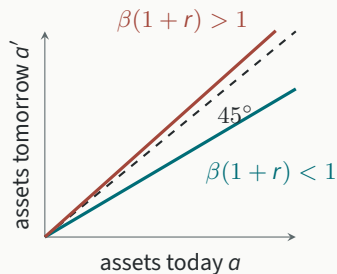
Six Steps, and the Spec Is the Contract



- ▶ A spec states four things: the assumptions and domain rules; the data or source restrictions; the procedure and its validation target; the outputs and where they are saved.
- ▶ Steps 1 to 3 produce no code. Ask the agent to restate the spec in its own words before it writes any.
- ▶ A specific request names its check: not “add some validation” but “an Euler-residual check, max and mean below 10^{-6} on the grid”.
- ▶ **The spec is the contract: the agent can be held to what it restated.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1–§4.2; Judd (1998).

Step 4: V0, and the Gate Written Before the Run; Steps 5 and 6



- ▶ **V0**: the simplest version with a benchmark you fully understand — at $\beta(1+r) = 1$ the policy is *on* the 45° line. The three cases and a tolerance enter the spec first; nothing is added until V0 passes.
- ▶ Steps 5 and 6 climb $V0 \rightarrow V1$ (+risk) $\rightarrow V2$ (+distribution) $\rightarrow V3$ (+prices); a broken rung sends you back to the last one that passed.
- ▶ A machine checks convergence, NaNs, determinism; you check limiting cases, signs, magnitudes, economic sense.
- ▶ **Session 4's lab ran it once: seven blocks, ten gates fixed in advance, none a loss value.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1 and §4.4; Session 4's lab notebook and its recorded gate file.

Grade the Plan Before Any Code: Five Criteria

1 it restates your goal, in its own words

2 steps you can review in minutes

3 a test per step, named in advance

4 stop and rollback points

5 the smallest testable object first

A plan without acceptance criteria is a wish. Send it back: a revision costs one message; a wrong build costs a day.

- ▶ Deck 6.2's plan mode shows you the plan; these five decide whether you approve it.
- ▶ In Part C, the first prompt applies the third criterion to the seed itself.
- ▶ **Approve against the criteria, not against the prose.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.2.

Validation Is Your Referee Checklist

workflow checks

convergence achieved
no NaN or Inf values
weights, mappings and source logs saved
residuals or quote checks pass
the script reproduces its output twice

economic checks

limiting cases recover known results
comparative statics make sense
magnitudes are plausible
inference uses the right errors and sample
you can say it in plain economics

- ▶ A machine can run the left column. Only you can run the right one.
- ▶ **If you cannot defend the output in a seminar, you are not done.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.4.

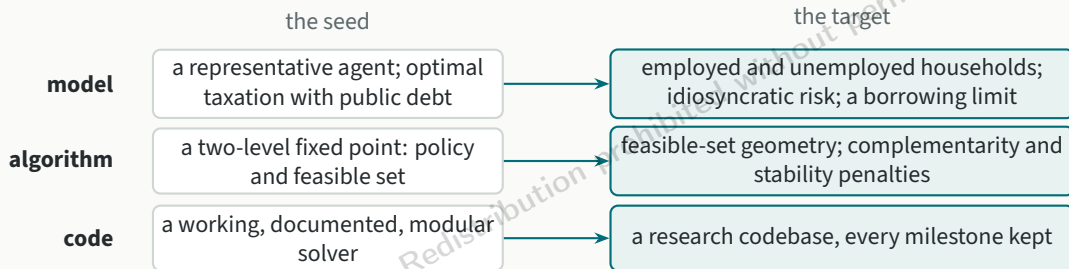
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A Months-Long Research Project, Built This Way

A validated seed, a team of sessions, a milestone ladder, three discoveries, and what stayed human



The Project: Where It Started, Where It Ended



- ▶ The seed was a working solver from an earlier paper; the target, an economy in which households need the debt to self-insure.
- ▶ **The agent compressed the loop between the three layers; it never chose the destination.**

Source: The instructor's research record (unpublished, coauthored), described by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

What Makes a Good Seed

it works

reproduces its own paper, not just runs

it is documented

a TeX note in the code's notation

it is modular

each run a config diff, not a fork

it points ahead

the no-risk case of the target

- ▶ The classroom V0 and the research seed play one role: the thing every later failure is measured against.
- ▶ Model, algorithm and code deform together. Keep the TeX, the pseudocode and the code describing one economy: drift between them is where silent errors live.
- ▶ **A good seed is to agentic research what a good instrument is to empirical work.**

Source: The instructor's research record, described by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

Opening the Conversation: Restate, Then Audit

1. "Here are the RA paper (TeX) and the RA repository.
Restate the model and the algorithm in your own words.
List every equation and where it is implemented."
2. "Audit the code against the document. Report every place
where the code and the TeX disagree: scoring, bounds,
sampling, penalties. Classify each as a correctness risk
or a documentation gap."
3. "Propose a refined RA document + code pair in which the
two are consistent. Do not add any new economics yet."

- ▶ Prompt 1 tests comprehension; prompt 2 tests the seed; prompt 3 fixes the foundation while freezing the economics. The audit's findings were kept as a file.
- ▶ **The audit is the cheapest insurance in the project: an inconsistent seed propagates into every later version.**

Source: The instructor's research record: the project's opening prompts, typed as shown (RA: representative agent), described by role.

The Division-of-Labour Contract

the agent

drafts and reorganises the LaTeX
turns equations into vectorised code
proposes penalty and boundary patterns
summarises diagnostics and mismatches

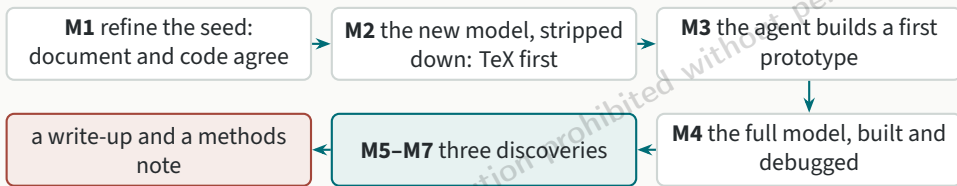
the economist

the economics of every formulation
which simplifications are acceptable
conceptual bugs: timing, direction
the benchmarks, and accepting results

- ▶ Written down at the start, before any code: who owns which kind of error.
- ▶ **The agent supplies throughput; the economist supplies judgement.**

Source: The instructor's research record: the project's division of labour, described by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

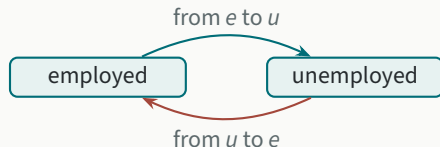
The Milestone Ladder, and Which Session Did What



- ▶ Separate sessions worked separate layers — model drafting, code review, debugging, documentation — all feeding one package.
- ▶ Each milestone was a validated artifact, kept; each failure of one version became explicit structure in the next.
- ▶ **The unit of progress is the validated milestone, not the week.**

Source: The instructor's research record, by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1 and §4.3.

Words Before Code: Two Bugs No Compiler Catches



$$a_{t+1} = (a_t + y_t - c_t) / q_t$$

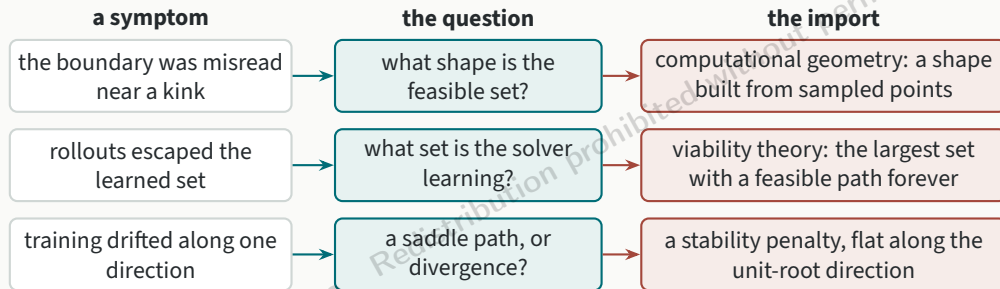
with c_t , not c_{t+1}

a assets, y income, c consumption, q
the bond price

- ▶ The agent drafted the stripped-down model's equations: syntactically fine, economically wrong in two places.
- ▶ The asset transition used next period's consumption; the budget constraint is a current-period restriction.
- ▶ The transition probabilities ran against the direction of movement between the employment states.
- ▶ **No compiler, test or residual catches a timing error. An economist reading the TeX does.**

Source: The instructor's research record: the stripped-down model, described by role.

Three Discoveries the Iteration Surfaced



- ▶ The second is the structure of self-generation in repeated games: the feasible set is the largest fixed point of a set operator.
- ▶ **Each came from iterating on why a run failed: the economist recognised the question; the agent carried the search.**

Source: The instructor's research record, described by role; Aubin (1991); Abreu, Pearce & Stacchetti (1990).

To Benefit the Most, You Need Four Things

a good seed

a working triple that points at the target

a clear vision

which problem is interesting, and why

good judgement

reject the plausible, wrong equation

orchestration

sessions on separate layers, one package

- ▶ What changed: the theory–code–debug loop compressed; documentation became executable; imports from other literatures got cheap.
- ▶ What did not is Deck 6.1's list: the AI never chose the question, never caught the timing on its own, never knew when a result was not credible.
- ▶ **The workflow is the product: it is the part you reuse on the next project.**

Source: The instructor's research record, by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1–§4.3.

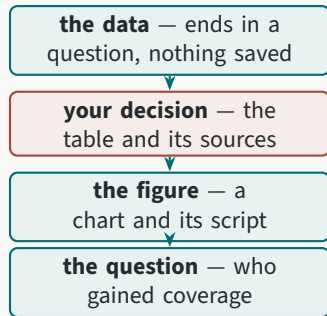
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The Checks That Stay Yours

What a clean run hides, how far the loop can run alone, and what to test before a number reaches the paper



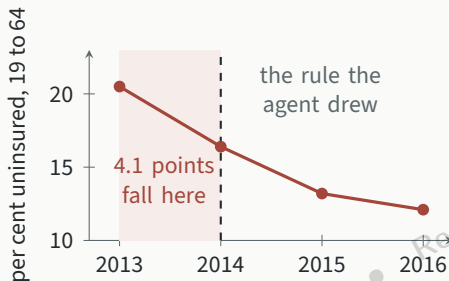
A Twenty-Two-Minute Task, in Four Prompts



- ▶ From an empty folder: four prompts, twenty-two minutes, recorded 13 September 2026 — uninsurance rates by age, 2008 to 2024, every source written down.
- ▶ The first ended with no data at all: the data service now wants a key issued to a person, so the agent stopped and asked.
- ▶ **The second prompt was not a correction. It was the judgement the agent lacked: one survey, and check the age groups before combining years.**

Source: Course-recorded agent run, 13 September 2026; U.S. Census Bureau, health-insurance historical tables (ACS).

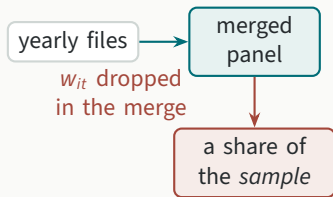
The Trap: An Instruction the Data Cannot Carry



- ▶ The fourth prompt asked for a January mark. Each point here is a one-year average sitting at its integer year.
- ▶ No position on this axis is a month, so the rule went at 2014 — the whole-year average, after the event.
- ▶ 4.1 of the 8.4-point fall, 48.8 per cent of it, ends up on the “before” side.
- ▶ **Every check passed. The instruction had no valid reading, and nothing in the loop says so.**

Source: Course-recorded agent run, 13 September 2026; U.S. Census Bureau, one-year estimates (ACS).

A Quiet Failure: The Lost Survey Weights



no error, no warning, every year wrong the same way ●

- ▶ The estimand is one line of arithmetic, the weighted uninsured rate

$$\hat{r}_t = \sum_i w_{it} \mathbf{1}\{\text{uninsured}\} / \sum_i w_{it}.$$

- ▶ Merging several annual files, an assistant lost the weight column: the pipeline reported the share of the *sample*, not the population rate.
- ▶ **Divergence announces itself; a pipeline error does not. This one surfaced at a group meeting.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.4; AHRQ, Medical Expenditure Panel Survey documentation.

Passed Every Gate, and Still Wrong: Our Own Numbers

Session 2

| 0.4%



Session 4

| 1.5%

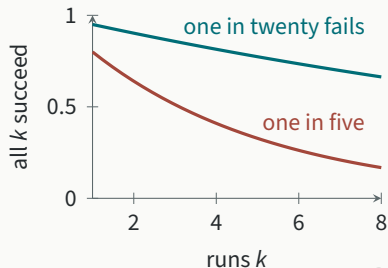


teal is the residual the check saw;
red is the error in aggregate capital

- ▶ Session 2, rung 1 from the default start: a held-out Euler residual of -2.38 on Judd's scale, the policy 25 per cent wrong, and capital at 0.06 against 28.35.
- ▶ Session 4, block D from a random start: a residual of -1.81 , capital 3.24 against the classical 5.64.
- ▶ **In both the machine check passed and the economics failed — which is why gate D reads capital within 5 per cent first, and the residual below -2.5 second.**

Source: Session 2's lab, block D (rung 1, default start); Session 4's lab, block D (random start) and its gate file.

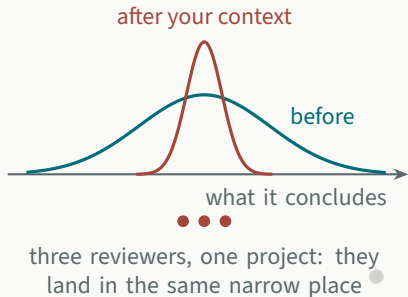
One Try Is Not a Measurement



- ▶ An agent left to run is a repeated trial: report the rate over k fresh runs, never the best run. pass^k , the chance all k succeed, falls fast even from high reliability.
- ▶ Published scores break two ways: leaked answers, and harness exploits that score well without solving anything — which is why no leaderboard number appears in this session.
- ▶ **Your own evaluation is a benchmark too: an agent that does nothing should score zero on it.**

Source: Yao et al. (2024), pass^k ; Miao & Feng, *AI for Economists*, ch. 15, §4.4.

The Context Dilemma, and the Prior Your Team Inherits



- ▶ Context acts as evidence: it narrows the output towards your framing, not the truth.
- ▶ Sharma et al.: sycophancy across frontier models — factual flip-flops, unwarranted concessions, matching the user's view.
- ▶ Perez et al.: it *scales with model size*; preference data reward the agreeable reply, so no “do not flatter me” brief line removes it.
- ▶ **Hint the theory “should” work and the model defends, not tests, the framing: keep one reader outside your context.**

Source: Xie et al., ICLR (2022); Ouyang et al. (2022); Sharma et al. (2023), arXiv:2310.13548; Perez et al. (2023), ACL Findings.

The Autonomy Slider, and Where Loops Converge

a machine decides

nobody but you

the tests pass

is the instrument
exogenous?

the benchmark
is matched

does the welfare
result hold?

the proof
checks in Lean

is the question
worth asking?

48 multiplications,
not more

is the result
believable?

left: the loop converges; right: it
drifts towards what it was last told

Source: Google DeepMind, AlphaEvolve (14 May 2025, a vendor account); Karpathy (2025); Miao & Feng, *AI for Economists*, ch. 15, §4.3.2; Deck 1.1's frame on the machine-checked proof.

- ▶ AlphaEvolve (DeepMind, May 2025): 4×4 complex matrices in 48 scalar multiplications, improving on Strassen (1969); a count decides, so the loop converged, as Deck 1.1's Navier–Stokes run did with a proof assistant.
- ▶ Set the autonomy slider per task: as far right as the criterion is decidable, and no further.
- ▶ Loop the decidable sub-tasks (tests, tuning, a benchmark it runs); keep the rest.
- ▶ **An agent that builds an agent stops where machines cannot decide right and wrong.**

Seven Tests Before It Goes into the Paper — and When Not to Delegate

test	the question
correctness	did it compute the object you meant; can you re-check magnitudes yourself
reproducibility	can a coauthor rerun a clean copy, with model, prompt, seed and data pinned
auditability	can a referee trace the commands, checks and sources behind it
privacy boundary	which files stayed local, and what content crossed the network
cost	what it cost in tokens and in money, including what a replicator will pay
speed	net of checking and repair, did it save time
skill investment	is your group building durable skill, or quietly losing it

- ▶ Keep a two-line edit or one-shot plot in the editor (no validation surface), and write the V0 gate before bringing in an agent: faster output is faster wrong.
- ▶ **Any “no” or “I do not know” means the output is not yet research-grade.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §5, the practice checklist, recast as tests of an output; the two boundary rules carried from the 2025 lecture on governance.

Five Habits, and What Deck 6.4 Adds

a system set up once

a brief, a team with a reviewer who did not build it, gates that refuse

restate the task
before any code

write the gate
before the run

grade the plan,
then approve

add one margin
at a time

keep one reader
outside your
context

- ▶ Each habit is cheap, and each is what one of this deck's two cases turned on.
- ▶ **Next, Deck 6.4: the one thing on your disk the agent still cannot read — your literature.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4 and §5.

Your Turn: Lab Blocks B and F, and the Design Canvas

block B — recompute all 96 numbers of the recorded run, then find the axis



block F — a gate that fires, one that never fires, and one agent run again



the canvas — eight cells to fill before an editor opens

- ▶ Block B is Part D's twenty-two-minute task, put under audit: the arithmetic reproduces; the axis does not.
- ▶ Block F runs Part A's rules as code. Self-paced, after class, offline; about forty minutes for both.
- ▶ **Then fill the canvas for a project of your own, before you open an editor.**

Source: The session's lab, blocks B and F, and its design canvas.